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This was due to thermal issues in particular. This brought about a major change in chipsets—Intel's very own 875/865 gave way to their 925/915 series. Recently the move to Core 2 Duos saw 975/965 chipsets emerge from 955/945 chipset based boards because of their incompatibility with the Core 2 Duo.

AMD has seen its share of changes and upheavals too. The move to DDR2 initiated by Intel to complement its then latest Pentium D processors forced AMD to shift to a new platform as well—AM2, which brought DDR2 compatibility to AMD systems.

With their latest generation of processors, AMD and Intel will be leaping to yet another platform. Intel has lots of 45nm plans based around Penryn—their Core 2 Duo successor. This involves a new chipset—code-named Bearlake, a.k.a. the P35, that will support these processors. This is the mid-range offering replacing the P965. There's support for DDR2 and DDR3 memory, 1333 MHz FSB support, a spanking new Southbridge, and more.

AMD also has a new socket for Barcelona—Socket F, which is a 1207-pin LGA-based affair; details are sketchy.

You have to ask yourself... are these new chipsets merely platforms to house faster processors? To a large extent, yes. For example, Intel's Bearlake is supposed to debut with native E-SATA support, and possibly a new generation of PCI-Express to provide further bandwidth to GPUs.

The notebook space has also seen a lot of upheaval with both Intel and AMD introducing new products late last month.

Intel's Santa Rosa (dubbed the Centrino Pro) is built around a 64-bit, dual-core Merom processor which should aid multitasking no end. There's inbuilt HSPDA (High Speed Downlink Packet Access)—a new protocol for mobile data transmission (also called 3.5G). There are other noticeable improvements too. Like Intel Active Management technology that allows improved remote system access, thereby solving many problems with reduced costs. Imagine disconnecting an infected notebook from a network remotely! Wireless Draft N support is included. Also in the package is integrated DX10 graphics.

Puma is AMD's code-name for their new notebook architecture and represents the platform as a whole. HyperTransport 3 connectivity and a new power saving DDR2 memory controller are the standouts here. The graphics will be a DX10 supporting core with full support for Blu-Ray and HD-DVD playback.

The latest generation of integrated graphics chipsets will be DX10 featuring unified shaders. Eventually DDR3 will become the Desktop standard—we expect greater bandwidth from the next generation PCI-Express standard to feed the latest and emerging GPUs and even multi-GPUs, that is, CrossFire and SLI configurations. ☐

Memory

With faster processors, memory manufacturers have to step up and make sure that their components aren't the weakest link in your PC

The memory market has seen its share of trials and tribulations the past couple of years. Now DDR2 is as common as cows on a highway, while DDR memory's availability is somewhat sporadic. 667 MHz (PC 5300) memory has replaced 533 MHz (PC 4200) memory as the new entry-level solution for Desktops. PC 6400 memory (800 MHz) DDR2 from brands like Corsair, Kingston, and Transcend are readily available. A 1 GB stick of DDR2 667 MHz memory can now be had for as little as Rs 2,000—512 MB of DDR memory costs the same, incidentally!

The enthusiast memory market is still in its infancy in India, with low-latency memory costing upward of Rs 15,000. Don't expect any miracles in this segment because demand for such memory is extremely low.

Intel promises DDR3 memory as a replacement for DDR2 over the next three years. The benefits of this would be lower power consumption and much higher bandwidth. The reason for the changeover is the speed barriers that DDR2 has hit (around 1.1 GHz), mainly due to thermal issues, and, of course, material limits. Intel visualises processor performance scaling higher, and they feel memory would bottleneck their quad-core offerings. Based on global sales,

